

Fresnel Biprism — II

B.Sc. (V Semester) — Physical Optics · Kumaon University

Unit II: Interference · Lecture 5 · Instructor **Dr. Prateek Sharma**

Experimental arrangement · Measuring d by the lens displacement method · Determination of wavelength

1 Experimental Arrangement

The slit S , the biprism, and the micrometer eyepiece are mounted on an optical bench. The eyepiece is focused on the fringes, and the fringe width β is measured by moving the cross-wire across a known number of fringes. The distance D is read from the bench.

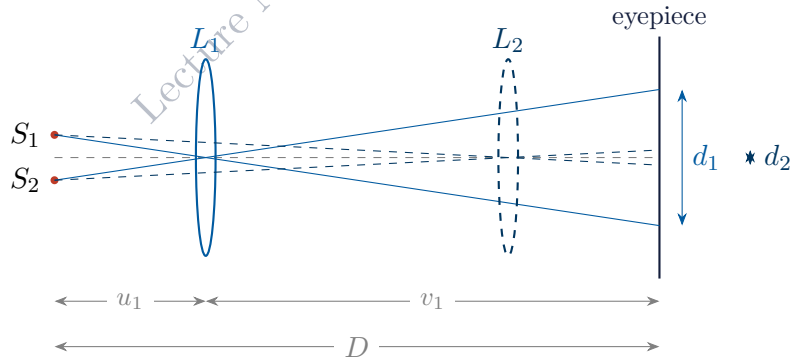
From Lecture 4,

$$\lambda = \frac{\beta d}{D},$$

so the only quantity still needed is the separation d of the virtual sources. It is found by the **lens displacement method**.

2 Measuring d : The Lens Displacement Method

A convex lens is placed between the biprism and the eyepiece. Keeping the object plane (the virtual sources) and the image plane (the eyepiece) fixed at a distance D apart, there are two lens positions L_1 and L_2 (provided $D > 4f$) at which the lens forms a sharp real image of S_1 and S_2 in the eyepiece.



(Only the rays through the optical centre are drawn. At L_1 the images are magnified; at L_2 they are diminished.)

Let the object separation be d . For lens position L_1 (object distance u_1 , image distance v_1), the linear magnification is

$$m_1 = \frac{v_1}{u_1},$$

so the measured image separation is

$$d_1 = m_1 d.$$

By the reciprocity of conjugate points, moving the lens to L_2 simply interchanges the object and image distances ($u_2 = v_1$, $v_2 = u_1$). Then its magnification is

$$m_2 = \frac{v_2}{u_2} = \frac{u_1}{v_1} = \frac{1}{m_1},$$

$$d_2 = m_2 d = \frac{d}{m_1}.$$

Multiplying,

$$d_1 d_2 = m_1 d \times \frac{d}{m_1} = d^2$$

$$d = \sqrt{d_1 d_2}$$

Note that the focal length of the lens is not needed: only the two image separations are measured.

3 Determination of Wavelength

Let $b_1 = v_1$ and $b_2 = u_1$. From the geometry,

$$u_1 = v_2 = b_2, \quad v_1 = u_2 = b_1,$$

$$D = u_1 + v_1 = b_1 + b_2.$$

Substituting $d = \sqrt{d_1 d_2}$ and $D = b_1 + b_2$ in $\lambda = \beta d / D$,

$$\lambda = \frac{\beta \sqrt{d_1 d_2}}{b_1 + b_2}$$